

Module 6: Field and particle physics

This module introduces the modern picture of fields and particle interactions as fundamental mechanisms of nature.

6.1 Fields

The idea of field has been met in the earlier module. The first section (6.1.1) treats the electromagnetic field in a practical context. The electric field, as the interaction between charges at rest, links back to the mathematically analogous model of the gravitational field. There are opportunities for discussing the social impact of the widespread distribution and use

of electrical power and its influence on industrial societies. (HSW9, HSW12).

The second section (6.1.2) covers interactions between charged particles and ideas about electric field and potential.

6.1.1 Electromagnetism

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) the action of a transformer: magnetic flux from a coil; induced e.m.f = rate of change of flux linkage	HSW2
(ii) the action of a dynamo: change of flux linked produced by relative motion of flux and conductor	
(iii) electromagnetic forces; qualitatively as arising from tendency of flux lines to contract or interaction of induced poles; quantitative calculation limited to force on a straight current-carrying wire in a uniform field	
(iv) simple linked electric and magnetic circuits: flux produced by current turns, need for large conductance and permeance and the effect of increasing the dimensions of an electromagnetic machine; qualitative effect of iron and air gap.	permeance is understood as a magnetic equivalent to electrical conductance and with analogous dependence on the dimensions and nature of the magnetic medium
(b) <i>Make appropriate use of:</i>	
(i) the terms: B -field, magnetic field, flux, flux linkage, induced e.m.f, eddy currents	
<i>by sketching and interpreting:</i>	
(ii) graphs of variations of currents, flux and induced e.m.f	
(iii) diagrams of lines of flux in magnetic circuits; continuity of lines of flux.	

(c) *Make calculations and estimates involving:*

(i) $\phi = BA, \mathcal{E} = -\frac{d(\phi N)}{dt}$

In the context of Faraday's and Lenz's laws
HSW1, 2, 8

(ii) $F = ILB$

Calculations restricted to current perpendicular to
uniform magnetic field

(iii) $\frac{V_1}{V_2} = \frac{N_1}{N_2}$ for an ideal transformer

i.e. no energy losses for an ideal transformer

Learners will also be expected to recall the equation
relating V and N

(iv) $\frac{I_2}{I_1} = \frac{N_1}{N_2}$ for an ideal transformer

Learners will also be expected to recall the equation
relating I and N
HSW12

(d) *Demonstrate and apply knowledge and
understanding of the following practical activities
(HSW4):*

(i) observing induced e.m.fs produced under
varying conditions such as dropping a
magnet through a coil attached to a data
logger or oscilloscope

links to **6.1.1a(ii)**, **6.1.1c(i)**

(ii) determining the uniform magnetic flux
density between the poles of a magnet
using a rigid current carrier and digital
balance.

links to **6.1.1a(iii)**, **c(ii)**

(iii) investigate transformers.

links to **6.1.1a(i)**, **c(iii)(iv)**